

PPAP STEP-BY-STEP GUIDE

7-Step PPAP Process | AIAG 4th Edition | IATF 16949 | Quality Head Agents

Step 1: Receive & Review Customer Requirements

- Obtain the customer drawing with latest revision and engineering change records
- Review Customer Specific Requirements (CSR) — TML, Maruti, Honda, Hyundai, Mahindra, AL
- Confirm submission level required (default: Level 3)
- Identify all CC (Critical) and SC (Significant) characteristics from drawing

Step 2: Complete Product Design Validation

- Issue/update DFMEA — cover all failure modes from customer drawing
- Verify design records match customer CAD/drawing revision exactly
- Complete engineering change documents if applicable
- Obtain Customer Engineering Approval (if required for drawing changes)

Step 3: Develop Process Documentation

- Create/update Process Flow Diagram (PFD) — all operations including incoming inspection
- Update PFMEA aligned to PFD — all new/changed operations covered
- Update Control Plan — all CC/SC characteristics with reaction plan
- Define and confirm all gauges and checking fixtures required

Step 4: Complete MSA (Measurement System Analysis)

- Identify all gauges used for CC/SC characteristics measurement
- Conduct Gauge R&R (GRR) study — minimum 10 parts × 3 operators × 2 trials
- Accept criteria: GRR < 10% excellent, 10-30% conditional, >30% reject
- Complete bias and linearity studies for key measurement systems

Step 5: Conduct Production Trial Run

- Produce MINIMUM 300 pieces using all production tooling, fixtures, operators
- Run at planned production rate (pieces/hour as stated on Control Plan)
- Collect dimensional results on 6 pieces — all balloon dimensions
- Calculate Cpk/Ppk — CC characteristics need $Cpk \geq 1.67$, others ≥ 1.33

Step 6: Compile PPAP Package

- Organize all 18 elements as per AIAG PPAP 4th Edition sequence
- Complete the Part Submission Warrant (PSW) — all fields mandatory
- Attach sample product (number per customer requirement)
- Have Quality Manager sign the PSW before submission

Step 7: Submit & Obtain Customer Approval

- Submit package to Customer Supplier Quality Engineer (SQE)
- Track submission — follow up within 5 working days if no acknowledgement

• Customer decision: Approved / Interim Approval / Rejected
• If rejected: implement corrective actions and resubmit immediately